

AFRICAN CRITICAL MINERALS INTELLIGENCE PLATFORM

Comprehensive Data Blueprint, Source Intelligence & Technical Architecture

**Bloomberg Terminal + GIS + Supply Chain Intelligence + AI
Surveillance for African Minerals**

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EXECUTIVE SUMMARY

The African Critical Minerals Intelligence Platform (ACMIP) is a next-generation mining monitoring and governance system that transforms fragmented, opaque African mineral operations into transparent, verifiable, real-time intelligence. It serves governments, mining operators, investors, and commodity markets by fusing multispectral satellite imagery, production and trade data, geospatial analytics, and AI-powered anomaly detection.

Unlike traditional exploration-focused platforms, ACMIP solves the harder operational challenge: knowing what is actually happening at African mine sites, in real time, when ground access is impossible or politically sensitive.

THE GOVERNANCE CRISIS IN NUMBERS	
\$89 billion Lost annually by Africa to illicit financial flows (UNCTAD / Africa Center)	\$1.3 trillion Left sub-Saharan Africa via IFFs since 1980, 60% from the resources sector (Greenpeace Africa)
\$250 billion Africa's critical mineral export revenues in 2023-2024 alone (Zawya)	\$16 trillion Projected 25-year revenues from just copper, nickel, cobalt & lithium in sub-Saharan Africa (WEF, 2026)

Core Value Propositions

- Operational Transparency:** Satellite-verified production activity vs. self-reported volumes — closing the reconciliation gap that costs African governments 10-25% of potential mining revenues annually.
- Regulatory Velocity:** Automated compliance dashboards replacing 6-month manual audit cycles with continuous, audit-ready ESG and production monitoring.
- Supply Chain Security:** Real-time critical minerals flow intelligence (lithium, cobalt, copper, graphite, rare earths) from African mine sites to global markets.
- Zero-Ground-Access Verification:** Remote monitoring of mine expansion, tailings dam stability, environmental encroachment, and artisanal mining incursion without site visits.
- Cost Efficiency:** \$300-\$1,500/month infrastructure cost vs. \$500K+ annual traditional consulting-led monitoring programs.

SECTION 1: CORE MINERAL CATEGORIES TO MONITOR

The platform classifies minerals into four strategic groups, each with distinct monitoring requirements, market dynamics, and governance priorities.

Mineral	Africa's Global Share	Top African Producers	Strategic Priority
Cobalt	54.5% of reserves	DRC (97.2% of African output)	CRITICAL — EV batteries
PGMs (Platinum, etc.)	79.3% of reserves	South Africa (71.5% of global platinum)	CRITICAL — Industry/green tech
Chromium	61.7% of reserves	South Africa (42.7% global)	HIGH — Stainless steel
Manganese	36.5% of reserves	South Africa, Gabon	HIGH — Steel, batteries
Copper	8.2% of reserves	DRC, Zambia, Namibia	CRITICAL — Energy transition
Lithium	5% of identified global resources (26.7Mt)	Zimbabwe, Mali, Namibia	CRITICAL — EV batteries
Gold	7.8% of reserves	Ghana (19.5% of Africa output), South Africa	HIGH — Financial, tech
Diamonds	32.4% of reserves	Botswana, DRC, Angola	HIGH — Industrial, gems
Bauxite	25.5% of reserves	Guinea (98.3% of African output)	HIGH — Aluminium
Iron Ore	Major producer	Guinea (Simandou), South Africa	HIGH — Steel
Rare Earth Elements	Significant deposits	Tanzania, Malawi, SA, Burundi	STRATEGIC — Tech supply chain
Graphite	Major deposits	Mozambique, Tanzania, Madagascar	STRATEGIC — EV batteries
Tantalum / Coltan	Significant	DRC, Rwanda	STRATEGIC — Electronics

Sources: USGS Mineral Commodity Summaries 2024/2025; Mining Technology (October 2025); African Energy Chamber (January 2026).

SECTION 2: PRIMARY DATA SOURCES

2A. Geological & Mine Location Data

USGS Africa Mineral GIS Geodatabase (Primary Foundation Layer)

The definitive starting point for the platform's geospatial backbone. In 2021, the USGS National Minerals Information Center (NMIC) published a comprehensive geodatabase covering all 58 mineral-producing African countries. A GeoPDF map update was released in July 2024.

Access: <https://doi.org/10.5066/P97EQWXP> | <https://pubs.usgs.gov/publication/ofr20241041/full>

The geodatabase contains 19 data layer categories:

Layer #	Data Layer Content
1	Mineral production and processing facilities (2,400+ operations)
2	Mineral exploration and development sites
3	Mineral occurrence sites and deposits (lat/lon with commodity, status, owner)
4-5	Undiscovered mineral resource tracts (potash, PGEs, copper, Gabon, Mauritania)
6	Coal occurrence areas
7-8	Electric power generating facilities and transmission lines
9-11	LNG terminals, oil/gas pipelines, hydrocarbon resource assessments
12	Cumulative production and recoverable conventional resources by nation
13	Major mineral-exporting maritime ports
14-15	Railroads and major roads
16-19	Major cities, lakes, river systems, administrative divisions (ADM1)

Key attribute fields per facility:

Field	Example	Field	Example
Mine Name	Kibali Gold Mine	Coordinates	Lat/Lon (decimal degrees)
Mineral Type	Gold	Operator	Barrick Gold
Status	Active / Closed / Exploration	Production Capacity	Tons per year

Ownership	Company + Government Share	Country Code	GENC 2-char alpha
Source Reference	USGS Minerals Yearbook Vol III	Facility Type	Mine / Processing / Brine

USGS Global Critical Minerals Deposits Database

A companion dataset covering global distribution of selected critical mineral mines, deposits, and districts. Covers: fluorine, gallium, hafnium, lithium, manganese, niobium, PGEs, REEs, rhenium, selenium, tantalum, tellurium, tin, titanium, vanadium, zirconium + 40 more commodities.

Access: <https://catalog.data.gov/dataset/global-distribution-of-selected-mines-deposits-and-districts-of-critical-minerals>

USGS World Minerals Exploration Database (WMED) — Unpublished

Compiled from company reports, press releases, scientific/geologic reports, industry and government reports. Highlights the general distribution of mineral exploration sites and potential future extraction areas. Accessible indirectly via the GIS Geodatabase export.

British Geological Survey (BGS) — World Mineral Statistics Dataset

Annual mineral production statistics for ~200 countries, with historical time series. Complements USGS data with independent verification.

Access: <https://nora.nerc.ac.uk/id/eprint/world-mineral-statistics> | Free download

S&P Global Market Intelligence / Wood Mackenzie

Commercial databases with detailed mine-level data including reserve/resource estimates, ownership structures, capital costs, and production forecasts. Required for institutional-grade analysis.

Attribute	Detail Available
Reserve/Resource Estimates	JORC/NI 43-101/SAMREC compliant figures
Ownership Structure	Parent company, subsidiaries, government equity stakes
Capital & Operating Costs	CAPEX/OPEX per tonne benchmarks
Production Forecasts	5-20 year mine-life projections
Permitting Status	License numbers, expiry, renewal flags

Open Street Map (OSM) + Natural Earth

Free, openly licensed geospatial base layers for roads, settlements, river systems, and administrative boundaries. Provides the geographic context layer underlying all mine-site visualizations. Access via Overpass API or bulk downloads.

2B. Production & Export Data

USGS Minerals Yearbook — Africa Volume (Volume III)

Annual country-by-country mineral production statistics. Published by the USGS National Minerals Information Center. Covers all extractable commodities per country, with operator names, production volumes, and export value estimates.

Access: <https://www.usgs.gov/centers/national-minerals-information-center/minerals-yearbook-metals-and-minerals>

USGS Mineral Commodity Summaries (Annual — Published January)

The most widely cited annual overview of global supply, demand, trade, reserves, and prices for 90+ mineral commodities. Contains country-level breakdown. 2024 edition data points:

- DRC: 74% of global cobalt mine production in 2023
- China: 67% of global refined cobalt production
- South Africa: 71.5% of global platinum, 42.7% of chromium in 2024
- Guinea: 98.3% of Africa's bauxite output in 2024
- Ghana: 19.5% of Africa's gold output in 2024

Access: <https://pubs.usgs.gov/periodicals/mcs2024/>

African Development Bank (AfDB) — African Natural Resources Centre (ANRC)

Publishes lithium-cobalt value chain analyses, mineral industrialization studies, and country-level production assessments. Key publication: 'Lithium-Cobalt Value Chain Analysis for Mineral Based Industrialization in Africa.'

Access: <https://africa-energy-portal.org> / <https://www.afdb.org/en/topics-and-sectors/sectors/energy>

World Bank — Extractives Data Portal

Hosts country-level extractive sector data, including cobalt market analyses (DRC-focused), mining governance reports, and fiscal regime data. Key resource: 'Cobalt in the Democratic Republic of Congo — Market Analysis.'

Access: <https://documents.worldbank.org>

Benchmark Mineral Intelligence — Cobalt Market Report

The most authoritative commercial source for cobalt supply intelligence. 2024 findings (cited from Cobalt Institute / Benchmark):

- CMOC produced 114kt of cobalt in 2024 — 31% above stated capacity at DRC operations
- CMOC's global cobalt market share rose from 24% (2023) to 41% (2024)
- Kisanfu (KFM): 50kt capacity; Tenke Fungurume (TFM): 37kt capacity
- Indonesia emerged as second-largest cobalt producer at 30kt+, up 82% year-on-year

Cobalt Institute — Annual Market Reports

Published annually. Covers global cobalt supply/demand, mine-level production, artisanal vs. industrial breakdown, pricing trends, battery sector demand.

Access: <https://www.cobaltinstitute.org/cobalt-market-report/>

Current African Critical Minerals Production Snapshot (2024-2025 Data)

Country	Mineral	2024 Output / Status	Key Operators
DRC	Cobalt	~190-200kt (97.2% of Africa)	CMOC, Glencore, Jinchuan
DRC	Copper	~2.5Mt; +12.8% extractive growth 2024	Ivanhoe (Kamoa-Kakula), CMOC, Glencore
South Africa	Platinum	~100t (71.5% of global)	Anglo Platinum, Impala, Sibanye
South Africa	Chromium	~19.7Mt (42.7% of global)	Samancor, Merafe, Tharisa
Ghana	Gold	19.5% of Africa's regional output 2024	Newmont, Gold Fields, Kinross
Guinea	Bauxite	~98.3% of Africa's total 2024	CBG, SMB-Winning, GAC
Guinea	Iron Ore	Simandou: 39.3Mt target by late 2025	Rio Tinto / Simfer, SMB-Winning
Zimbabwe	Lithium	Leads African LCE output; 10.8Mt Bikita reserve	Prospect Lithium, Zhejiang Huayou
Zambia	Copper	+19.2% forecast 2025 (937.5kt)	ZCCM (Mopani), First Quantum
Morocco	Phosphate	World's largest reserves (70%+)	OCP Group
Namibia	Uranium	Major global supplier	Rössing (CGNPC), Husab (SWAKOP)
Burkina Faso	Gold	5th largest Africa producer; fast-growing ASM	Endeavour Mining, government

Sources: Mining Technology (Oct 2025), African Energy Chamber (Jan 2026), USGS MCS 2024, US Commercial Service DRC (Feb 2026).

2C. Trade & Customs Intelligence

UN Comtrade Database (Primary Trade Data Source)

The world's most comprehensive international trade database. Over 1 billion trade records from 1962. Covers 140+ reporter countries with annual and monthly import/export statistics by HS commodity code.

Feature	Details
Coverage	Annual + monthly data; 200 countries; 99%+ of world merchandise trade
Commodity System	HS (Harmonized System) codes — search minerals by HS chapter 26 (ores/slag/ash)
Free API Access	500 API calls/day; 100,000 records/query with free registration
Premium API	Unlimited commercial access via UN Comtrade+ subscription
Key Use Case	Compare self-reported African export data vs. partner-country import data to detect misinvoicing
Access URL	https://comtradeplus.un.org / https://comtrade.un.org/labs/data-explorer/
HS Codes for Minerals	Chapter 26: ores; 2601-2621 covers iron, copper, nickel, cobalt, aluminium, chromium, manganese, etc.

UNCTAD Trade Analysis Information System (TRAINS)

Combines trade data with customs tariffs and non-tariff measures (NTMs). Covers 170+ countries with HS-based tariff data. Useful for identifying export restrictions, duties, and trade barriers affecting African mineral exports.

Access via World Integrated Trade Solution (WITS): <https://wits.worldbank.org> | UNCTAD Data Hub: <https://unctadstat.unctad.org>

ITC Trade Map (International Trade Centre)

Joint ITC/WTO/UN product. Provides bilateral trade flow data, market analysis, and supply/demand trends by commodity. Highly useful for African minerals bilateral corridor analysis (e.g., DRC→China copper flows).

Access: <https://www.trademap.org>

UNCTAD SDG Pulse — Critical Minerals Trade Dashboard (2025)

Published June 2025. Key findings on African mineral trade:

- Africa nearly doubled copper exports in the reporting period (+98.2% to \$44 billion)
- DRC-China bilateral mineral corridor exceeds \$10 billion annually
- Africa records steady growth in energy-transition-related mineral exports
- Africa's critical minerals identified as high trade opportunity but high supply-risk for Europe

Access: <https://sdgpulse.unctad.org/critical-minerals/>

EITI — Extractive Industries Transparency Initiative Data Portal

The EITI works with 28 African countries (out of 51 global implementing countries). Publishes annual reconciliation reports comparing company payments vs. government receipts across license allocation, production, revenues, state participation, contracts, and beneficial ownership.

EITI Data Element	Monitoring Value
License Allocations	Who holds exploration/mining rights, when granted, how long
Production Data	Country-level and company-level production volumes by commodity
Revenue Streams	Royalties, taxes, dividends, bonuses — by company and by government entity
State Participation	Government equity stakes in mining operations
Beneficial Ownership	Underlying ownership structures behind mining companies
Contract Transparency	Mining agreements, fiscal terms, community obligations
ASM Data	Artisanal and small-scale mining registrations (Ghana, Tanzania emphasis)
Emissions Data	GHG emissions disclosures now mandated by 2023 EITI Standard revision
2024 Progress Report	https://eiti.org/eiti-progress-report-2024

EITI Progress Report 2024 African Country Scores (Selected):

Country	EITI Score	Status	Key Notes
Madagascar	67/100	Active (Feb 2024)	Revenue reporting gaps; limited civil society engagement
Mali	67/100	Active (Jul 2024)	Contract transparency progress; political instability risks
Cameroon	53/100	Active (Feb 2024)	Data reliability and timeliness issues
Central African Republic	45/100	Active (Nov 2024)	Severe challenges due to political instability and insecurity

Global Financial Integrity (GFI) — Trade Misinvoicing Analysis

GFI conducts systematic analysis of under-invoicing and over-invoicing in African mineral exports by comparing UN Comtrade bilateral data. Key finding from Burkina Faso EITI study: \$4.93 billion in IFFs over 2012-2021, with gold alone accounting for 61%, detected via trade misinvoicing. The UAE was confirmed as a key destination for gold of fraudulent origin.

Source: <https://eiti.org/data-use-case/burkina-faso-quantifying-illicit-financial-flows-mining>

2D. Satellite & Geospatial Monitoring

Free/Open Satellite Data Sources

Satellite / Platform	Key Specifications & Access
ESA Copernicus Sentinel-2 (Optical)	Twin satellites; 10m-60m resolution; 13-band multispectral; 3-5 day revisit; FREE. Ideal for mine expansion, deforestation, tailings, vegetation change. Access: Copernicus Open Access Hub, AWS (registry.opendata.aws/deafrica-sentinel-2)
ESA Copernicus Sentinel-1 (SAR Radar)	C-band SAR; ~20m resolution; 12-day revisit; penetrates cloud cover; FREE. Critical for Africa (high cloud coverage). Detects surface mining scars, water bodies, infrastructure. Access: AWS (registry.opendata.aws/deafrica-sentinel-1)
NASA/USGS Landsat 8 & 9	30m resolution; 16-day revisit; free since 1972. Strong historical baseline (50+ years) for change detection. Access: USGS Earth Explorer, Google Earth Engine
Digital Earth Africa (DE Africa)	Continental-scale cloud-free composites (GeoMAD), Water Observations from Space (WOfS), Cropland Map, Change Detection. Preprocessed, analysis-ready Sentinel-1/2 and Landsat data for all Africa. Access: https://digitalearthafrika.org AWS Open Data Registry
ESA Pléiades (via Third Party Mission)	Sub-metre resolution (0.5m); used by researchers at Univ. of Padova to detect ASM in Burkina Faso semi-desertic environments; commercial license required
GMES & Africa Programme	EU-African Union partnership; 13 consortia, 144 institutions, 45 African countries; provides 1.2 petabytes/year of EO data via Sentinel satellites for natural resource monitoring

Commercial Satellite Providers

Provider	Resolution & Features
Planet Labs (PlanetScope)	3m daily imagery; near-daily global coverage. Ideal for operational monitoring. API available. Commercial pricing per km2.
Maxar (WorldView-3/4)	0.3m resolution; historical archive. Gold standard for detailed site verification.
Airbus Defence & Space (SPOT, Pléiades)	0.5m-6m resolution; reliable archive. European alternative to US providers.
SAR Data (Capella Space, ICEYE)	Sub-metre SAR; near-daily; penetrates clouds/night. Emerging provider with Africa relevance.
Swift Geospatial (Africa-focused)	Remote sensing specifically for African mining operations. ESG auditing, before-and-after restoration verification, illegal mine detection. https://swiftgeospatial.solutions
Farmonaut	Multispectral satellite monitoring for African mine sites; AI analytics; blockchain traceability integration for gold/platinum supply chains. https://farmonaut.com

AI & Deep Learning Mine Detection Research

Validated academic methodologies directly applicable to the platform:

Research / Method	Details & Citation
U-Net CNN on Sentinel-2 (West Africa)	Mapped ASM locations across 1.75M km ² in 13 Sub-Tropical West African countries. Achieved pixel-level mining classification. (PMC, 2022)
Multispectral CNN — Ghana ASM Detection	CNN on open-source Sentinel-2 imagery; distinguishes mining, built/developed, and vegetation land use classes. Low-cost scalable monitoring system. (ScienceDirect, 2020)
Deep Learning Mine + Tailings Dam Detection	Uses Google Earth Engine + Sentinel-2 + TensorFlow; discovered 263 unregistered mines in Brazil. Directly transferable methodology to Africa. (arXiv, 2020)
SAR-based Dredge Detection (CAR)	USGS used SAR to map diamond/gold mining dredges on the Kadéï River, CAR from 2015-2019 using 387 SAR scenes. (USGS Data Release, 2021)
ADSMS U-Net (Burkina Faso)	ESA-backed study using Pléiades + Sentinel-2 to detect ASM in semi-desertic environments; fewer false negatives. (ESA EO Gateway, 2023)
Earthrise Mining Detector (Open Source)	Neural network ensemble detecting artisanal gold mines in Sentinel-2 imagery; 99.6% precision; quarterly updates from Q2 2025. https://github.com/earthrise-media/mining-detector

Processing Platforms & APIs

- Google Earth Engine (GEE): Cloud-based planetary-scale geospatial analysis; free for research; \$\$ for commercial. Petabytes of satellite data + JavaScript/Python API.
- AWS Open Data Registry: Hosts DE Africa Sentinel-1/2 products as free STAC-compliant datasets.
- STAC (SpatioTemporal Asset Catalog): Open standard for indexing satellite imagery; all DE Africa products are STAC-compliant.
- Open Data Cube (ODC): Open-source Python library for working with geospatial raster data; underpins Digital Earth Africa.
- ArcGIS / ESRI: Commercial GIS platform; widely used in African governments for mining cadastre and licensing.

2E. Supply Chain Traceability

OECD Due Diligence Guidance for Responsible Supply Chains

The OECD's flagship framework for 3TG (tin, tungsten, tantalum, gold) and cobalt supply chain due diligence from conflict-affected and high-risk areas. Adopted by EU regulation and referenced by SEC Section 1502 (Dodd-Frank). Mandatory for companies using African 3TG minerals.

Access: <https://www.oecd.org/corporate/mne/mining.htm>

iTSCI (ITSCI Tin Supply Chain Initiative)

The dominant mineral traceability scheme for 3TG in the Great Lakes Region of Africa (DRC, Rwanda, Burundi, Uganda). Bag-and-tag system that tracks minerals from mine site to export. Membership-based; data shared with implementing governments.

Better Cobalt / Responsible Cobalt Initiative (RCI)

Industry-led programs for cobalt supply chain verification in DRC. Better Cobalt (Cobalt Institute-managed) conducts site assessments, community benefit mapping, and blockchain-based certification. RCI led by Chinese companies (CNIA) addresses ASM practices.

Fairmined / Fairtrade Gold Standards

Certification schemes for artisanal and small-scale gold miners. Generate verifiable chain-of-custody documentation applicable to ASM operations in West Africa (Ghana, Burkina Faso, Guinea).

Blockchain-Based Traceability Platforms

Platform	Application to African Minerals
IBM Food Trust / TradeLens (blockchain logistics)	Precedent model; commodity tracking from origin to buyer
Minespider	Blockchain-based battery minerals traceability; DRC cobalt focus
Sourcemap	Supply chain mapping and transparency; artisanal mining
RCS Global / Better Sourcing Program	On-the-ground verification + digital chain-of-custody for cobalt and 3TG
Circular	EV battery supply chain traceability; cobalt/lithium from Africa
Farmonaut Blockchain	Satellite-verified, fraud-resistant supply chains for African gold/platinum

2F. Maritime & Port Intelligence

AIS (Automatic Identification System) Vessel Tracking

The foundational layer for monitoring mineral export flows from African ports to global smelters and refineries. AIS data enables: commodity vessel identification, voyage route tracking, port call analysis, dark vessel detection (AIS-off), and sanctions screening.

AIS Data Provider	Key Capabilities
Kpler (Commercial)	13,000+ AIS receivers; terrestrial + LEO satellite; <5 sec average latency; <1 min satellite latency; 10+ year historical archive; API/Snowflake/FTP delivery. Maritime + commodity trade intelligence combined. https://www.kpler.com/product/maritime
MarineTraffic (Commercial)	World's largest marine traffic tracking network; vessel history, port calls, cargo records. API available.
VesselFinder (Free/Commercial)	Basic real-time AIS tracking; limited historical. Good for proof-of-concept.
OECD AIS Tracking Dashboard (Free)	Country-level indicators on vessel activity, capacity, trade estimates, efficiency, and product-level commodity flows. https://www.oecd.org/en/data/dashboards/monitoring-maritime-trade
Windward Maritime AI	AI-powered maritime risk intelligence; sanctions compliance, dark vessel detection, OFAC compliance screening. Critical for detecting illicit mineral shipments.
UNODC / UN Global Platform	AIS-based trade monitoring; validated by major crisis events (Suez blockage, Red Sea attacks)

Key African Mineral Export Ports

Port	Country	Primary Minerals Exported	Annual Capacity
Port of Dar es Salaam	Tanzania	Gold, coal, copper, graphite	Major East Africa hub
Port of Mombasa	Kenya	Regional corridor; DRC/Uganda minerals	East Africa gateway
Durban / Richards Bay	South Africa	Coal, manganese, chrome, PGMs	Largest in Africa
Port Elizabeth (Gqeberha)	South Africa	Manganese, chrome	Dedicated manganese terminal
Port of Matadi / Boma	DRC	Copper, cobalt, coltan	Primary DRC export port
Port of Lobito (Lobito Corridor)	Angola	Copper/cobalt transit from DRC/Zambia	Strategic expansion 2024-2026
Port of Conakry	Guinea	Bauxite, iron ore	Largest bauxite port in Africa
Port of Tema	Ghana	Gold, bauxite, manganese	West Africa's largest port
Port of Lagos / Apapa	Nigeria	Coal, iron ore, solid minerals	Largest port by volume
Port of Toamasina	Madagascar	Nickel, cobalt, ilmenite	Eastern Africa mineral hub

2G. ESG & Compliance Data

ESG Compliance Market Overview

The global ESG compliance in mining market was valued at \$4.53 billion in 2024 and is projected to reach \$9.55 billion by 2033 (CAGR 8.9%). The Middle East & Africa segment generated \$151.4 million in 2024, growing at 8.2% CAGR. (Grand View Research, 2025)

Key ESG Data Sources & Frameworks

Source / Framework	Data Available & Access
EITI Standard (2023 revision)	Now includes emissions disclosures as core requirement. 28 African countries implementing. Annual reconciliation reports with production, revenue, contract, and beneficial ownership data.
ICMM (International Council on Mining & Metals)	Voluntary ESG reporting framework; 27 major mining members including Glencore, Anglo American, Barrick. Annual sustainability reports.
Responsible Minerals Initiative (RMI / RMAP)	Smelter/refiner audit programs for 3TG and cobalt. Audit results publicly disclosed. Critical for supply chain due diligence.
GRI Standards (Global Reporting Initiative)	International ESG reporting standard; used by listed mining companies for sustainability disclosures.
IRMA (Initiative for Responsible Mining Assurance)	Third-party mine-site verification against social/environmental standards. Public audit results.
CDP (Carbon Disclosure Project)	Climate, water, and forest risk disclosures from listed mining companies. API access available.
Bureau Veritas Africa	Third-party ESG assurance for mines; sustainability report verification; audit and certification services. https://www.bureauveritas.africa/mine-esg-services
NRGI Resource Governance Index	Country-level governance quality scores for extractive sectors. 80+ countries assessed. https://resourcegovernance.org
African Union — Africa Green Minerals Strategy (2025)	Continental framework endorsed in 2025 for sustainable mineral development. Policy compliance benchmark.

Environmental Monitoring Indicators (Satellite-Derived)

Indicator	Measurement Method
Deforestation around mine sites	Sentinel-2 NDVI change detection; Landsat time series
Tailings dam extent and stability	SAR InSAR (surface deformation); optical area mapping
Water body turbidity / contamination	Sentinel-2 water quality indices (NDWI, turbidity)
Dust and particulate emissions	MODIS aerosol optical depth; Sentinel-5P tropospheric NO2

Night-time light activity	VIIRS/DMSP NTL; proxy for operational activity levels
Land surface temperature	Landsat thermal; proxy for industrial heat signatures
Mine pit expansion	DEM (Digital Elevation Model) differencing; Sentinel-1 SAR
Artisanal mining encroachment	U-Net CNN change detection on Sentinel-2 time series
Rehabilitation / revegetation	NDVI trend analysis; before-and-after composites

2H. Real-Time Market Analytics

Data Source	Coverage & Access
LME (London Metal Exchange)	Real-time spot and futures prices for copper, nickel, cobalt, aluminium, lead, zinc, tin. API via data vendors (Refinitiv, Bloomberg). https://www.lme.com
Benchmark Mineral Intelligence	Cobalt, lithium, nickel, graphite price assessments; supply/demand forecasts; battery metals intelligence. Commercial subscription.
Fastmarkets (formerly Metal Bulletin)	Daily price assessments for base metals, precious metals, battery materials. Cobalt hydroxide/sulfate spot prices. Commercial.
Reuters / Refinitiv Eikon	Real-time commodity prices, news flow, company filings. API integration. Commercial.
Bloomberg Terminal	ECTR function (bilateral trade flows); AHOY function (commodity shipments between ports); full mineral price suite. Commercial.
World Bank Commodity Price Data (Pink Sheet)	Monthly commodity price data; free download. Copper, cobalt, gold, bauxite, phosphate, iron ore benchmarks. https://www.worldbank.org/en/research/commodity-markets
IMF Primary Commodity Prices	Monthly commodity price time series; free API. https://www.imf.org/en/Research/commodity-prices
Statista	Aggregate production statistics from multiple sources for quick reference; requires subscription for downloads.

2I. Conflict, Sanctions & Political Risk Data

Source	Data Type & Access
ACLED (Armed Conflict Location & Event Data Project)	Real-time conflict event data; point-level; country/region/actor breakdowns. Free API. Critical for DRC, Mali, Burkina Faso, CAR mining zones. https://acleddata.com
Global Witness	Investigative reports on conflict minerals, beneficial ownership, human rights abuses in African mining. https://www.globalwitness.org
UN Panel of Experts Reports (DRC, CAR, Sudan, Mali)	UN Security Council-mandated expert panels document illicit mineral flows, armed group financing, and sanctions violations. Free download from UN.
OFAC Sanctions List (US Treasury)	Specially Designated Nationals (SDN) list. API access. Cross-reference against mine ownership and trading entity beneficial owners.
EU Conflict Minerals Regulation	Due diligence requirements for 3TG imports into EU. Compliance data available for EU-registered importers.
Global Transparency Initiative (OpenOwnership)	Beneficial ownership registers; cross-reference mining license holders against PEP/sanctions lists. https://www.openownership.org
KPMG / EY Mining Country Risk Reports	Annual country-level investment risk, regulatory change, and political stability assessments for African mining jurisdictions.

SECTION 3: AI ANOMALY DETECTION ARCHITECTURE

The AI layer is what transforms the platform from a data aggregator into an intelligence system. It detects discrepancies, flags risks, and surfaces insights that no manual process could capture at scale or speed.

3A. Anomaly Detection Use Cases

Anomaly Type	Detection Method & Data Inputs
Production-Export Mismatch	Cross-reference EITI production reports vs. UN Comtrade export volumes vs. AIS vessel cargo declarations. Statistical outlier flagging per mine/country/quarter.
Undeclared Mining Activity	Sentinel-2/SAR change detection at licensed claim boundaries; U-Net CNN identifies new mine scars outside registered polygons.
Tailings Dam Instability	InSAR surface deformation analysis (1-5mm precision); SAR coherence change; alerts if dam wall displacement exceeds threshold.
Satellite Night-Light Spikes	VIIRS night-time light anomaly detection at mine site coordinates; proxy for unauthorized or unreported night operations.
Trade Misinvoicing Detection	Statistical comparison of African country export value vs. partner-country import value for same HS code shipment. >15% divergence triggers alert.
Vessel Dark Activity Near African Ports	AIS gap analysis (vessel transponder off >X hours near mineral export port); correlate with known mineral shipment corridors.
Unauthorized ASM Expansion	U-Net CNN sequential classification of Sentinel-2 composites; detects pit expansion, vegetation loss, water diversion in and around licensed concessions.
Beneficial Ownership Red Flags	Graph analysis of company ownership structures; identify PEP (politically exposed persons), sanctioned entities, and opaque offshore holders in mining license chains.
Environmental Encroachment	NDVI degradation analysis within protected area buffers around mine sites; flag biodiversity corridor intrusions.

3B. Machine Learning Models

- Convolutional Neural Networks (CNN / U-Net): For pixel-level mine detection and land use classification from satellite imagery. Validated at 99.6% precision (Earthrise, 2025) and on Sentinel-2 for ASM across 13 West African countries.
- Time Series Anomaly Detection (LSTM / Isolation Forest): Applied to production volumes, export values, and vessel movement data to detect statistical outliers over time.

- Graph Neural Networks (GNN): For ownership structure analysis — mapping corporate hierarchies to detect hidden beneficial owners, circular ownership, and shell company patterns.
- Random Forest / XGBoost Classifiers: For ESG risk scoring combining satellite, financial, governance, and conflict data into composite risk indices.
- NLP (Named Entity Recognition + Sentiment): For processing news feeds, company reports, and regulatory filings to extract mine names, production figures, and risk signals automatically.

3C. Processing Infrastructure

Component	Technology Options
Geospatial Processing	Google Earth Engine (GEE), AWS SageMaker Geospatial, Open Data Cube (ODC)
Database Layer	PostGIS (spatial SQL), MongoDB (document store), TimescaleDB (time-series)
Data Pipeline	Apache Kafka (real-time streaming), Apache Airflow (batch orchestration)
ML Training	AWS SageMaker, Google Vertex AI, Azure ML
Visualization	Mapbox GL JS, Leaflet.js, Deck.gl, Kepler.gl (Uber open-source)
API Gateway	FastAPI (Python), GraphQL, REST JSON
Alert System	Webhooks, Email, Slack/Teams integration; threshold-based and ML-based triggers
Infrastructure Cost	\$300-\$1,500/month (satellite data: free Sentinel; cloud compute: per-use)

SECTION 4: GOVERNANCE GAPS THE PLATFORM ADDRESSES

4A. The Revenue Leakage Problem

Evidence	Source
\$89 billion lost annually to illicit financial flows	UNCTAD / Africa Center (2026)
\$1.3 trillion left sub-Saharan Africa via IFFs since 1980; 60% from resources sector	Greenpeace Africa (Oct 2025)
\$40 billion of IFFs from gold sector alone	AFRODAD / UNCTAD data
DRC lost \$1.2B in 5 mining asset deals 2010-2012 through underpricing	AFRODAD Report (2023)
Guinea mining tax expenditures = 24.4% of total national tax expenditure (2019)	IMF / WEF (Apr 2026)
Sierra Leone: \$131M in tax holidays to 2 firms = entire 2024 budget deficit	WEF (Apr 2026)
Burkina Faso: \$4.93B IFFs in gold & zinc 2012-2021 via trade misinvoicing	EITI / ITIE Burkina Faso (Jul 2025)
Uganda: IFFs enabled by informal mining, corruption, weak mineral database management	GFI / ACODE (Jan 2025)

4B. Artisanal & Small-Scale Mining (ASM) Monitoring Gap

ASM is particularly difficult to monitor yet represents a critical governance frontier:

- ASM employs approximately 12.5 million people in DRC alone
- ASM historically accounted for ~10% of DRC's cobalt output (2020); share has fallen to ~2% as large-scale mining expanded, but absolute volumes remain significant
- ASM gold smuggling rife across SADC due to porous borders and lack of formalization
- UNODC runs capacity-building programs in 50+ countries for satellite-based illegal mining monitoring
- SAR imagery (Sentinel-1 + USGS) successfully detected riverine dredges on the Kadei River, CAR, 2015-2019

4C. The Audit Cycle Problem

Traditional monitoring is episodic, expensive, and easily gamed:

- Manual audit cycles: 6-12 months in many African jurisdictions
- Physical site visits: Impossible or politically sensitive in conflict zones (eastern DRC, Mali, CAR, Burkina Faso)
- Traditional consulting-led monitoring: \$500,000+ per year per mine cluster
- ACMIP provides continuous monitoring at \$300-\$1,500/month infrastructure cost

4D. Supply Chain Opacity

Problem	Platform Solution
Self-reported production figures unverified	Satellite-derived activity indicators cross-referenced against declared volumes
Export declarations understate value/volume	UN Comtrade bilateral comparison (export vs. partner-country import data)
Beneficial ownership opacity in license holders	Graph-based ownership analysis linked to OpenOwnership/EITI registers
Artisanal mining flows invisible to customs	Satellite detection of ASM sites + bag-and-tag traceability scheme integration
No real-time port-level export intelligence	AIS vessel tracking + port call analysis at African mineral export ports
Tailings dam failures undetected until collapse	InSAR deformation monitoring + SAR coherence alerts

SECTION 5: PLATFORM INTELLIGENCE QUESTIONS & DATA SOURCES

This section maps each core intelligence question to the specific data sources, APIs, and methodologies that answer it.

Intelligence Question	Primary Data Source(s) & Method
Where is every major mineral deposit in Africa?	USGS Africa GIS Geodatabase (Layer 3 — deposits); USGS Global Critical Minerals Database; BGS World Mineral Statistics
Which mines are active right now?	Sentinel-2 NDVI/activity change detection; VIIRS night-light; USGS Minerals Yearbook status codes; Planet Labs daily imagery
Who owns them?	USGS GIS (owner attribute); EITI beneficial ownership disclosures; S&P Global / Wood Mackenzie; OpenOwnership register
What is current production volume?	EITI annual production disclosures; USGS MCS/Yearbook; company sustainability reports; satellite-derived activity proxy
Where are minerals exported?	UN Comtrade HS Chapter 26 bilateral flows; UNCTAD TRAINS; AIS vessel cargo correlations; ITC Trade Map
Which ports and smelters are used?	USGS GIS Layer 13 (mineral export ports); AIS port call analysis (Kpler/MarineTraffic); USGS processing facilities data
Are there illegal mining patterns?	Sentinel-2/SAR U-Net CNN beyond licensed polygons; UNODC satellite monitoring; Digital Earth Africa change detection; night-light spikes
Are production figures consistent with exports?	EITI reconciliation reports; UN Comtrade bilateral mirror comparison; GFI trade misinvoicing analysis
Are there sanctions, conflict, or ESG risks?	ACLED conflict data; OFAC SDN list; UN Panel of Experts reports; Global Witness investigations; EITI scores
Which countries dominate specific minerals?	USGS MCS 2024/2025; AfDB ANRC; African Energy Chamber Critical Minerals Report (Jan 2026); UNCTAD SDG Pulse (Jun 2025)
What are future reserve depletion forecasts?	S&P Global / Wood Mackenzie mine-life models; USGS undiscovered resource tracts (GIS Layers 4-5); Benchmark Mineral Intelligence forecasts
Is a tailings dam stable?	InSAR deformation monitoring (Sentinel-1); SAR coherence analysis; NDVI around dam; historical displacement records
Is a vessel transporting undeclared minerals?	AIS dark-vessel detection (Windward); cargo manifest vs. declared HS codes; port call departure/arrival cross-reference with known mineral corridors

SECTION 6: IMPLEMENTATION ROADMAP

Phase 1 — Foundation (Months 1-3)

- Ingest USGS Africa GIS Geodatabase into PostGIS spatial database
- Set up Sentinel-2 and Sentinel-1 automated ingestion via Digital Earth Africa API
- Build UN Comtrade trade data pipeline (HS Chapter 26 minerals, 54 African reporter countries)
- Create EITI data parser for 28 African implementing countries (production, revenue, license data)
- Deploy base map visualization with mine location, status, owner, commodity layers

Phase 2 — Intelligence Layer (Months 4-6)

- Train U-Net CNN on Sentinel-2 composites for mine activity detection (baseline from open-source models)
- Build production-vs-export reconciliation engine using UN Comtrade bilateral mirror data
- Integrate ACLED conflict data with mine location layer (spatial join, 25km buffer alerts)
- Add OFAC sanctions screening for mine ownership entities
- Deploy night-time light (VIIRS) anomaly detection for unreported operational activity

Phase 3 — Real-Time & Commercial (Months 7-12)

- Integrate AIS vessel tracking API (Kpler or MarineTraffic) for port-level mineral export monitoring
- Add commercial pricing data (LME, Benchmark Mineral Intelligence) for market context layer
- Build ASM incursion detection system using SAR change detection + licensed concession polygons
- Deploy tailings dam monitoring using InSAR deformation alerts (Sentinel-1)
- Launch automated EITI compliance dashboard replacing manual audit cycle
- Productize: API, dashboard, alert system for government/investor clients

SECTION 7: KEY REFERENCES & SOURCE DIRECTORY

Official Data Portals

Source	URL
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USGS Africa GIS Geodatabase	https://doi.org/10.5066/P97EQWXP
USGS GeoPDF Map (July 2024)	https://pubs.usgs.gov/publication/ofr20241041/full
USGS Mineral Commodity Summaries 2024	https://pubs.usgs.gov/periodicals/mcs2024/
USGS Africa Programs	https://www.usgs.gov/international-programs/africa
Digital Earth Africa	https://digitalearthafrica.org
DE Africa AWS Open Data	https://registry.opendata.aws/collab/deafrica/
Copernicus Open Access Hub	https://scihub.copernicus.eu
ESA EO Africa Programme	https://eo4society.esa.int/eo-africa/
UN Comtrade+	https://comtradeplus.un.org
UNCTAD SDG Pulse — Minerals	https://sdgpulse.unctad.org/critical-minerals/
ITC Trade Map	https://www.trademap.org
World Bank Extractives	https://documents.worldbank.org
EITI Data Portal	https://eiti.org
EITI Progress Report 2024	https://eiti.org/eiti-progress-report-2024
NRGI Resource Governance Index	https://resourcegovernance.org
ACLED Conflict Data	https://acleddata.com
OpenOwnership Register	https://www.openownership.org
World Bank Commodity Prices	https://www.worldbank.org/en/research/commodity-markets
Cobalt Institute Market Report	https://www.cobaltinstitute.org/cobalt-market-report/
UNODC Illegal Mining Monitoring	https://www.unodc.org/unodc/en/data-and-analysis/statistics/crime/monitoring-illegal-mining.html

Key Research Publications

Publication	Citation
USGS Africa GIS Data Release	Padilla et al. (2021), U.S. Geological Survey data release, doi:10.5066/P97EQWXP
CNN ASM Detection — Ghana	Maconachie et al. (2020), ScienceDirect, doi:10.1016/j.rse.2020.111917
ASM Mapping West Africa	PMC (2022), U-Net on Sentinel-2, 1.75M km ² coverage
Tailings Dam Detection Brazil	Balaniuk et al. (2020), arXiv:2007.01076
SAR Dredge Detection — CAR	Alessi et al., USGS Data Release (2021)
ASM Detection Burkina Faso	Nava et al., ESA EO Gateway (Feb 2023)
Cobalt Market Report 2024	Benchmark Mineral Intelligence / Cobalt Institute (May 2025)

Africa Critical Minerals Report	African Energy Chamber (January 2026)
UNCTAD Critical Minerals Trade	UNCTAD SDG Pulse (June 2025), UN Comtrade basis
Illicit Financial Flows — Greenpeace	Greenpeace Africa (October 2025)
WEF — Africa Critical Minerals	World Economic Forum (April 2026)
IFF Burkina Faso Mining Study	ITIE Burkina Faso / EITI (July 2025)
Uganda Mining IFF Report	GFI / ACODE (January 2025)
Africa Mining Sector Outlook 2025	Mining Technology (October 2025)
Lithium-Cobalt Value Chain Africa	AfDB ANRC, accessible at africa-energy-portal.org

African Critical Minerals Intelligence Platform — Data Blueprint v1.0 | May 2026

Compiled using data from: USGS, AfDB, EITI, UN Comtrade, UNCTAD, World Bank, Cobalt Institute, Digital Earth Africa, ESA, ACLED, Mining Technology, African Energy Chamber